

Occupation-Level Forecast Methodology

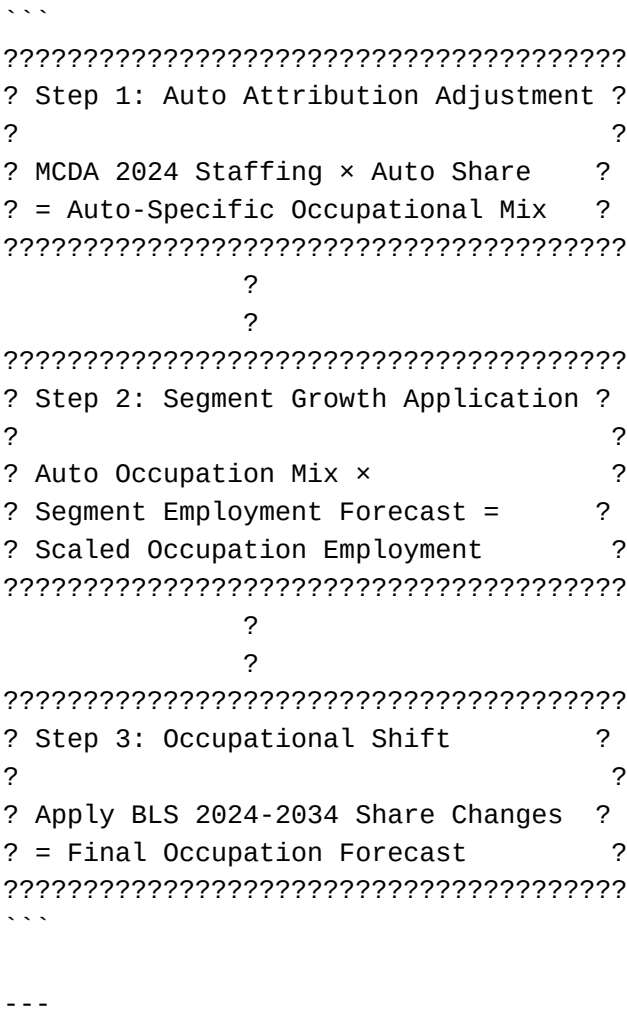
Overview

This document describes the methodology for generating occupation-level employment forecasts for Michigan's automotive supply chain segments from 2024 through 2034, with particular emphasis on 2030 projections for workforce planning.

Analytical Framework

Three-Step Methodology

The occupation forecasting approach combines three distinct analytical steps, each addressing a specific dimension of workforce projection:



Step 1: Auto Attribution Adjustment

Objective

Isolate the automotive-specific portion of 2024 MCDA staffing patterns by discounting employment in industries that serve both automotive and non-automotive markets.

Data Inputs

- **MCDA 2024 Staffing Patterns** (`data/interim/mcda\_staffing\_long\_2021\_2024.csv`)
- Occupation-level employment by segment for 2024
- Based on Michigan OEWS data aggregated to 10 automotive segments

- ****BEA Auto Attribution Shares**** (`data/raw/auto\_attribution\_bea.csv`)
 - Defines proportion of each NAICS code attributable to automotive sector
 - Based on Bureau of Economic Analysis industry definitions
- ****Lightcast Auto Attribution Shares****
 (`data/raw/auto_attribution_core_auto_lightcast.csv`)
 - Alternative definition from Lightcast Core Auto research
 - Industry-specific research on automotive exposure

Methodology

1. ****Compute Segment-Level Auto Shares****:
 - For each segment, calculate employment-weighted average auto share across constituent NAICS codes
 - Formula: $\text{segment_auto_share} = \frac{\sum(\text{naics_employment} \times \text{naics_auto_share})}{\sum(\text{naics_employment})}$
2. ****Apply to Occupational Employment****:
 - For each occupation within each segment:
 - $\text{auto_occupation_employment} = \text{mcda_employment} \times \text{segment_auto_share}$
 - Maintains proportional occupational mix (assumes auto and non-auto portions have similar staffing)
3. ****Preserve Both Attribution Approaches****:
 - Generate separate datasets for BEA and Lightcast attributions
 - Enables sensitivity analysis on attribution methodology choice

Key Assumption

****Occupational mix is similar between auto and non-auto portions of each industry****. For example, if NAICS 3261 (Plastics Manufacturing) is 31% automotive, we assume the occupational distribution is similar in the automotive 31% as in the total industry.

****Limitation****: May not hold for highly specialized industries where auto work requires different skill sets. However, data constraints prevent more granular estimation.

Example Calculation

Segment 1 (Materials & Processing):

- NAICS 3261: Employment = 34,371, BEA Share = 0.608, Lightcast Share = 0.310
- NAICS 3262: Employment = 5,147, BEA Share = 0.608, Lightcast Share = 0.670
- ...additional NAICS codes

Weighted BEA Share = $(34,371 \times 0.608 + 5,147 \times 0.608 + \dots) / 72,030 = 0.45$

Weighted Lightcast Share = $(34,371 \times 0.310 + 5,147 \times 0.670 + \dots) / 72,030 = 0.38$

For SOC 51-4041 (Machinists) with 2024 MCDA employment = 1,200:

- BEA-adjusted employment = $1,200 \times 0.45 = 540$
- Lightcast-adjusted employment = $1,200 \times 0.38 = 456$

Step 2: Segment Growth Application

Objective

Scale auto-specific occupational employment from 2024 through 2034 using segment-level employment forecasts.

Data Inputs

- ****Segment Employment Forecasts**** (4 files in `data/processed/`):
 - `mi\_qcew\_segment\_employment\_timeseries\_bea\_extended\_moody.csv` - BEA attribution, Moody's growth
 - `mi\_qcew\_segment\_employment\_timeseries\_bea\_extended\_bls.csv` - BEA attribution, BLS growth
 - `mi\_qcew\_segment\_employment\_timeseries\_coreauto\_extended\_moody.csv` - Lightcast attribution, Moody's growth
 - `mi\_qcew\_segment\_employment\_timeseries\_coreauto\_extended\_bls.csv` - Lightcast attribution, BLS growth

Methodology

1. ****Match Attribution Approaches****:
 - BEA-adjusted occupations use BEA segment forecasts
 - Lightcast-adjusted occupations use Lightcast segment forecasts
2. ****Apply Segment Growth****:
 - For each occupation-segment-year combination:
 - Retrieve segment total employment for that year
 - Without Step 3 adjustments: $\text{occupation_employment_year} = \text{base_2024_employment} \times (\text{segment_employment_year} / \text{segment_employment_2024})$
3. ****Preserve Methodology Combinations****:
 - 2 attribution approaches $\times$ 2 growth sources = 4 forecast scenarios
 - Enables sensitivity analysis on both attribution and growth rate assumptions

Example Calculation

Continuing the Machinists example:

Segment 1 forecast (BEA $\times$ Moody's):

- 2024: 32,000 total employment
- 2030: 29,500 total employment (7.8% decline)

Machinists forecast (BEA $\times$ Moody's), without occupational shift:

- 2030: $540 \times (29,500 / 32,000) = 498$

Step 3: Occupational Shift Incorporation

Objective

Account for changing occupational composition within segments over time (e.g., increasing engineers, declining production workers).

Data Inputs

- ****BLS Industry $\times$ Occupation Matrices**** (`data/raw/us\_staffing\_patterns/\*.csv`)
 - Occupation shares within each NAICS code for 2024 and 2034
 - From BLS Employment Projections Table 1.9
 - Detailed (6-digit SOC) occupational granularity

Methodology

- Compile BLS Occupational Shares**:
 - For each NAICS code in segment framework:
 - Extract "2024 Percent of Industry" and "Projected 2034 Percent of Industry" for each occupation
 - Aggregate to segment level using QCEW employment weights
- Interpolate Intermediate Years**:
 - Linear interpolation between 2024 and 2034 shares
 - Formula for year t: $\text{share}(t) = \text{share}_{2024} + (\text{share}_{2034} - \text{share}_{2024}) \times ((t - 2024) / 10)$
- Apply Occupational Shares**:
 - For each occupation-segment-year:
 - $\text{occupation_employment} = \text{segment_employment} \times \text{bls_share}(\text{year})$
 - Overrides constant-share assumption from Step 2
- Fallback Logic**:
 - If BLS data unavailable for occupation-NAICS pair: maintain 2024 share constant
 - Tracks which occupations have BLS shift data vs. constant share (`has_bls_shift` flag)

Key Assumption

BLS national occupational trends apply to Michigan automotive segments. Michigan's shift from production to engineering roles may differ from US averages, but lack of MI-specific projections necessitates this assumption.

Example Calculation

Machinists in Segment 1:

- BLS 2024 share: 1.75% of segment employment
- BLS 2034 share: 1.50% of segment employment (declining due to automation)

For 2030 (6 years into 10-year projection):

- Interpolated share = $1.75\% + (1.50\% - 1.75\%) \times (6/10) = 1.60\%$

Segment 1 total employment (BEA × Moody's) in 2030: 29,500

Machinists 2030 forecast (final):

- $29,500 \times 0.0160 = 472$ (vs. 498 without occupational shift adjustment)

Methodology Combinations

The analysis produces **4 distinct forecast scenarios**:

Scenario	Attribution	Growth Source	Use Case
BEA × Moody's	BEA auto shares	Moody's sector forecasts	Government definition, macro trends
BEA × BLS	BEA auto shares	BLS industry projections	Government definition, detailed industry

| ****Lightcast × Moody's**** | Lightcast Core Auto | Moody's sector forecasts | Commercial research, macro trends |
| ****Lightcast × BLS**** | Lightcast Core Auto | BLS industry projections | Commercial research, detailed industry |

This 2×2 matrix enables sensitivity analysis on:

1. ****Auto attribution uncertainty****: How sensitive are results to different definitions of "automotive industry"?
2. ****Growth rate uncertainty****: Do Moody's macro trends differ from BLS detailed projections?

Key Outputs

Primary Datasets

1. ****Full Time Series**** (`mi\_occupation\_employment\_forecasts\_2024\_2034.csv`)
 - Every occupation × segment × year × methodology combination
 - Columns: `segment\_id`, `segment\_name`, `occupation\_code`, `occupation\_title`, `year`, `employment`, `attribution`, `growth\_source`, `methodology`, `has\_bls\_shift`
2. ****2030 Snapshot**** (`mi\_occupation\_employment\_forecast\_2030.csv`)
 - Filtered to year 2030 (priority for workforce planning)
 - Same structure as full time series
3. ****2030 Summary Report**** (`mi\_occupation\_2030\_summary\_report.csv`)
 - Occupation-level aggregation across methodologies
 - Columns: `occupation\_code`, `occupation\_title`, `min\_employment`, `max\_employment`, `avg\_employment`, `std\_employment`, `segment\_name`
 - Shows methodology spread for each occupation

Separate by Attribution

- ****BEA Forecasts**** (`mi\_occupation\_employment\_forecasts\_bea\_2024\_2034.csv`)
- ****Lightcast Forecasts**** (`mi\_occupation\_employment\_forecasts\_lightcast\_2024\_2034.csv`)

Data Quality & Limitations

BLS Occupational Shift Coverage

- ****BLS data available****: ~60-70% of occupation-segment pairs
- ****Constant share fallback****: Remaining 30-40% use 2024 share through 2034
- ****Impact****: Underestimates occupational change in segments/occupations without BLS data

Attribution Assumptions

- ****Same occupational mix****: Assumes auto and non-auto portions have similar staffing
- ****Reality****: May vary (e.g., auto plastics might have more quality inspectors than food packaging plastics)
- ****Mitigation****: Two attribution approaches provide bounds

National vs. Michigan Trends

- **BLS projections**: Based on US national industry trends
- **Michigan reality**: State-specific factors (policy, major plant closures, supplier concentration) not captured
- **Mitigation**: Segment-level Michigan forecasts (Step 2) localize the analysis

Temporal Interpolation

- **Linear assumption**: Occupational share changes assumed linear 2024-2034
- **Reality**: May be non-linear (e.g., sudden automation events, regulatory changes)
- **Impact**: Mid-decade projections (2027-2030) most affected

Out-of-Scope Factors

This methodology does **not** explicitly model:

- Occupational migration patterns (workers moving in/out of state)
- Retirement and demographic effects
- Educational pipeline constraints (engineering program capacity)
- Skill upgrading within occupations (e.g., machinists learning CNC programming)
- Firm-specific workforce strategies

These factors should be considered when interpreting results for policy decisions.

Validation & Quality Checks

Internal Consistency Checks

1. **Occupation totals sum to segment totals** (within rounding tolerance)
2. **2024 base year matches MCDA × auto share** for each methodology
3. **Methodology spread is reasonable** (typically 5-15% range for 2030)

External Validation

Recommended comparisons:

1. **Segment 7 (Core Automotive)** 2030 forecast vs. OEM announced capacity plans
2. **Engineering occupations** growth vs. electrification workforce needs literature
3. **Production occupations** decline vs. automation adoption rates in auto manufacturing

Plausibility Heuristics

- Total 2030 employment should be within $\pm 20\%$ of 2024 (extreme scenarios only)
- No single occupation should dominate segment ($>40\%$ share)
- Top 5 occupations should account for 30-50% of segment employment
- Growth/decline rates should align with segment trends (not opposite direction)

Usage Guidelines

For Workforce Planning

- **Focus on 2030 snapshot** for near-term planning horizon
- **Use average across methodologies** as central estimate
- **Use min/max range** for scenario planning
- **Flag occupations with large methodology spread** for additional research

For Policy Analysis

- **Compare BEA vs. Lightcast** to assess sensitivity to auto definition

- ****Examine occupations with vs. without BLS shift data**** to identify data gaps
- ****Track top growing/declining occupations**** for training program targeting

For Stakeholder Communication

- ****Lead with 2030 average estimates**** (single number for clarity)
- ****Show methodology range**** (min-max) to convey uncertainty
- ****Highlight top 10-15 occupations**** by employment or growth
- ****Explain Step 1-2-3 logic**** in non-technical language

Technical Implementation

Script

`scripts/create\_occupation\_forecasts.py`

****Runtime****: ~2-5 minutes (depending on system)

****Dependencies****: pandas, numpy

****Inputs**** (read from `data/` subdirectories):

- MCDA staffing patterns
- Attribution shares (BEA and Lightcast)
- Segment forecasts (4 files)
- BLS occupational shift data (30+ files)
- Segment assignments lookup

****Outputs**** (written to `data/processed/`):

- 3 main CSV files
- 2 attribution-specific CSV files
- 1 summary statistics file

Notebook

`notebooks/14\_occupation\_forecasts.ipynb`

****Purpose****: Interactive analysis, visualization, and quality checks

****Sections****:

1. Load and preview forecast data
2. Top occupations analysis (2030)
3. Growth/decline analysis (2024-2030)
4. Segment-level aggregations
5. Methodology comparison
6. Time series visualizations
7. Occupational shift impact
8. Export summary report

References

Data Sources

- ****MCDA Staffing Patterns****: Michigan Center for Data and Analytics, derived from BLS OEWS

- **BLS Employment Projections**: 2024-2034 projections, released 2024
- **BEA Attribution**: Bureau of Economic Analysis auto industry definitions
- **Lightcast Attribution**: Core Auto research, industry-specific exposure analysis

Methodological Precedents

This approach adapts techniques from:

- BLS Employment Projections methodology (occupational staffing patterns)
- Regional economic impact modeling (industry-specific employment multipliers)
- Workforce development planning (occupation-level forecasting for training programs)

Related Documentation

- methodology.md - Decision log with dated entries
- data_sources.md - Full data provenance
- ../notebooks/README.md - Notebook 14 description
- ../scripts/README.md - Script documentation (to be created)

Updates & Revisions

Version 1.0 (2025-10-02): Initial methodology documentation

Future revisions should note:

- Changes to attribution approaches
- New data sources added
- Methodological refinements
- Validation findings that necessitate adjustments

For questions about this methodology:

1. Review executed notebook `14\_occupation\_forecasts\_executed.ipynb` for worked examples
2. Examine script comments in `create\_occupation\_forecasts.py` for implementation details
3. Check methodology decision log in `methodology.md` for context on choices made